EAST WEST UNIVERSITY

Project report on,

‘Pharmacy Shop Management System’

Course Title: Advanced Database Systems

Course Code: CSE464
Section: 01

Submitted to:

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Description:

Pharmacy Shop Management System Project is computer-based software developed for managing, monitoring and recording various activities of medical stores. It plays a key role in improving the productivity of medical stores through automated functions. It also helps to overcome the manual pharmacy management issues. Pharmacy Shop Management System Project will play a key role in minimizing human strain and errors when it comes to pharmacy management.

- Easy management of pharmacy
- Managing staff information and activity
- Managing financial details
- Maintain medicines details
- Data consistency
- Overcome the hassle of transaction
- Easy information refreshing
- Backup information

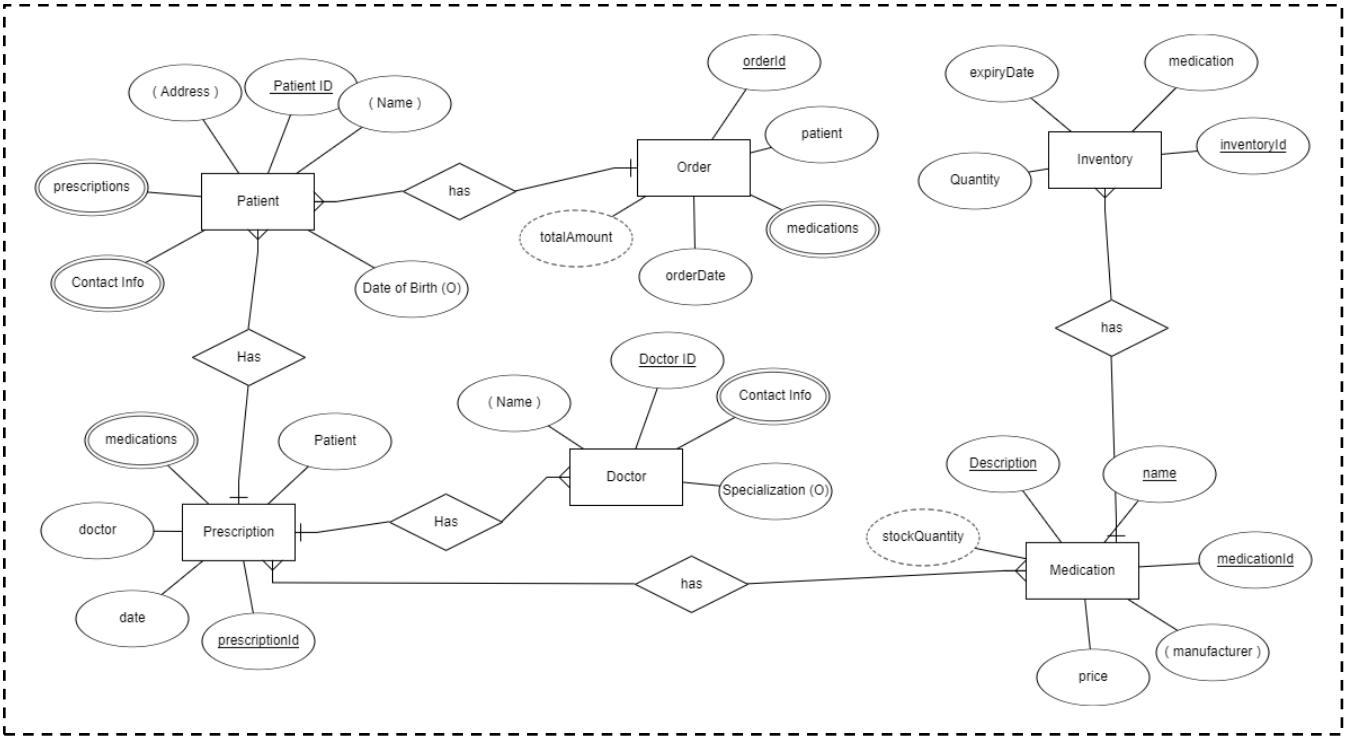
Pharmacy Shop Management System will manage records of all medicines, price details of medicines land, the number of medicines sold, brand details of medicines, staff at medical store, customer, transaction and other important details. The shop has some shelves, and these shelves consist of different categories of medicine. The database maintains the record. It deals with recording and processing data so that the medical store owners can easily manage the operations. Pharmacy Shop Management System Project is additionally used to produce reports of overall productivity of pharmacy. Moreover, it will be helpful in data processing and improving financial matters of pharmacy. Pharmacy Shop Management System Project is being used in a lot of areas as it has enormous applications. Using this system the pharmacy shop owners would get rid of manual time-consuming activities related to management.

Objective:

Pharmacy shop Management System database to help pharmacists to manage pharmacy in systematic ways. Pharmacy Management System can make the work easier by giving the details of medicine when its name is entered. A computer gives the details of the medicine like rate of medicine, and expiry date of the medicine. It becomes very difficult in big medical stores to handle the details of all the medicines manually, so by using this pharmacy management system We can maintain the records of all the medicines. It is fed with information whenever new medicines are brought and it is provided. search option. When we write the name of a medicine it gives the details of the medicine. A manual system we must search a medicine in a shelf depending on its category or the first name of this medicine but if we maintain a database then it is easy to search in a shelf. It reduces our hassle.

The technology platform in implementing this system uses Erdplus for E-R diagram and ORACLE APEX for SQL databases.

E-R Diagram:



SQL Schema:

```
CREATE OR REPLACE TYPE PhoneNumberList AS TABLE OF VARCHAR2(15);
/

CREATE OR REPLACE TYPE Name AS OBJECT (
  first_name VARCHAR2(50),
  last_name VARCHAR2(50)
);
/
CREATE OR REPLACE TYPE MedicationOrder AS OBJECT (
    medicationId VARCHAR2(10),
    quantity NUMBER
);
/
CREATE OR REPLACE TYPE MedicationRef AS OBJECT (
    medicationId VARCHAR2(10),
    quantity NUMBER
);
/
```

```
CREATE OR REPLACE TYPE MedicationList AS TABLE OF MedicationRef;
/
CREATE TYPE MedicationOrderList AS TABLE OF MedicationOrder;
/
CREATE OR REPLACE TYPE Address AS OBJECT (
   street VARCHAR2(100),
   city VARCHAR2(50),
   state VARCHAR2(50),
   zipCode VARCHAR2(10)
);
/
CREATE OR REPLACE TYPE ContactInfo AS OBJECT (
   phoneNumbers PhoneNumberList,
   email VARCHAR2(50)
);
/
CREATE TABLE Patient (
   patientId VARCHAR2(10) PRIMARY KEY,
   name Name, -- Composite
   dateOfBirth DATE NULL, -- Optional
   address Address NULL, -- Composite and Optional
   contactInfo ContactInfo NULL -- Composite and Optional
) NESTED TABLE contactInfo.phoneNumbers STORE AS patient_phone_numbers_nt;
/

CREATE TABLE Doctor (
   doctorId VARCHAR2(10) PRIMARY KEY,
   name Name, -- Composite
   specialization VARCHAR2(50) NULL, -- Optional
   contactInfo ContactInfo NULL -- Composite and Optional
) NESTED TABLE contactInfo.phoneNumbers STORE AS doctor_phone_numbers_nt;
/
CREATE TABLE Medication (
   medicationId VARCHAR2(10) PRIMARY KEY,
   name VARCHAR2(50) UNIQUE,
   description VARCHAR2(255) NULL, -- Optional
   manufacturer VARCHAR2(50) NULL, -- Optional
   price NUMBER (10, 2)
);
/
```

```
CREATE TABLE Inventory (
   inventoryId VARCHAR2(10) PRIMARY KEY,
   medicationId VARCHAR2(10) REFERENCES Medication(medicationId),
   quantity NUMBER,
   expiryDate DATE
);
/
CREATE TABLE Prescription (
   prescriptionId VARCHAR2(10) PRIMARY KEY,
   pdate DATE,
   doctorId VARCHAR2(10) REFERENCES Doctor(doctorId),
   patientId VARCHAR2(10) REFERENCES Patient(patientId),
   medications MedicationList -- Multivalued (nested table)
) NESTED TABLE medications STORE AS medications_nt;
/
CREATE TABLE OrderTable (
   orderId VARCHAR2(10) PRIMARY KEY,
   patientId VARCHAR2(10) REFERENCES Patient(patientId),
   orderDate DATE,
   status VARCHAR2(20),
   medications MedicationOrderList
) NESTED TABLE medications STORE AS order_medications_nt;

/
CREATE TABLE Inventory (
   inventoryId VARCHAR2(10) PRIMARY KEY,
   medicationId VARCHAR2(10) REFERENCES Medication(medicationId),
   quantity NUMBER,
   expiryDate DATE
);

/
CREATE VIEW MedicationWithStock AS
SELECT
   m.medicationId,
   m.name,
   m.description,
   m.manufacturer,
   m.price,
   COALESCE(SUM(i.quantity), 0) AS stockQuantity
```

```
FROM
   Medication m
LEFT JOIN
   Inventory i ON m.medicationId = i.medicationId
GROUP BY
   m.medicationId,
   m.name,
   m.description,
   m.manufacturer,
   m.price;
/
CREATE VIEW OrderWithTotalAmount AS
SELECT
   o.orderId,
   o.patientId,
   o.orderDate,
   o.status,
   (
      SELECT COALESCE(SUM(m.price * om.quantity), 0)
      FROM TABLE(o.medications) om
      JOIN Medication m ON om.medicationId = m.medicationId
   ) AS totalAmount
FROM
   OrderTable o;
```

Demo Data :

1. Insert value in Doctor Table:

```
INSERT INTO Doctor (doctorId, name, specialization, contactInfo)
VALUES (
   'D01',  Name('Dr.Arafat','Azim'),  'Neurology',
   ContactInfo(
      PhoneNumberList('01783457892'),
      'dr.arafat@gmail.com'
   )
);
```

See demo Data from Doctor table:

```
SELECT
    d.doctorId,d.name.first_name, d.name.last_name,d.specialization,
    p1.COLUMN_VALUE AS phoneNumber,d.ContactInfo.email
FROM
    Doctor d, TABLE(d.contactInfo.phoneNumbers) p1;
```

DOCTORID	NAME.FIRST_NAME	NAME.LAST_NAME	SPECIALIZATION	PHONENUMBER	CONTACTINFO.EMAIL
D05	Dr. Md. Mahmudul	Hasan	Neurology	01673939321	dr.mahmudul@gmail.com
D05	Dr. Md. Mahmudul	Hasan	Neurology	01783643283	dr.mahmudul@gmail.com
D04	Dr.Mofizur	Rahman	Medicine	01805893932	dr.rahmani@gmail.com
D04	Dr.Mofizur	Rahman	Medicine	01783647823	dr.rahmani@gmail.com
D10	Dr. wasif	Chisty	Urologist	01667834562	dr.wasif.com
D10	Dr. wasif	Chisty	Urologist	01784647364	dr.wasif.com
D06	Dr. Md. Rakibul	Hasan	Urologist	01873939421	dr.rakibul@gmail.com
D06	Dr. Md. Rakibul	Hasan	Urologist	01783563282	dr.rakibul@gmail.com
D09	Dr. Sami	Ahmed	Medicine	01673935678	prof.drsami.com
D09	Dr. Sami	Ahmed	Medicine	01784689364	prof.drsami.com

More than 10 rows available. Increase rows selector to view more rows.

2. Insert demo data in Patient table:

```
INSERT INTO Patient (patientId, name,  dateOfBirth, address, contactInfo
) VALUES (
    'P01', Name('Anis','Rahman'), TO_DATE('1980-05-15', 'YYYY-MM-DD'),
    Address('Aftabnagar', 'Dhaka', 'Bangladesh', '1212'),
    ContactInfo(
        PhoneNumberList('01834672348', '01456789234'),  'anisrahman@.com'
    )
);
```

See demo data from patient table:

```
SELECT
    p.patientId,p.name.first_name AS first_name,p.name.last_name AS last_name,
    p.dateOfBirth,p.address.street AS street, p.address.city AS city,
    p.address.state AS state,p.address.zipCode AS zipCode,
    t1.COLUMN_VALUE AS phoneNumber,p.contactInfo.email AS email
FROM
    Patient p, TABLE(p.contactInfo.phoneNumbers) t1;
```

Results Explain Describe Saved SQL History

PATIENTID	FIRST_NAME	LAST_NAME	DATEOFBIRTH	STREET	CITY	STATE	ZIPCODE	PHONENUMBER	EMAIL
P01	Anis	Rahman	05/15/1980	Aftabnagar	Dhaka	Banglades	1212	01834672348	anisrahman@.com
P01	Anis	Rahman	05/15/1980	Aftabnagar	Dhaka	Banglades	1212	01456789234	anisrahman@.com
P06	Rakib	Hosain	03/12/2004	Mohammadpur	Khulna	Bangladesh	1990	01745674574	rakib@.com
P06	Rakib	Hosain	03/12/2004	Mohammadpur	Khulna	Bangladesh	1990	01654567834	rakib@.com
P04	Zafor	Ahmed	03/12/1995	Uttra	Dhaka	Bangladesh	1212	01745785674	zafor@.com
P04	Zafor	Ahmed	03/12/1995	Uttra	Dhaka	Bangladesh	1212	01756787834	zafor@.com
P09	saikat	Biswas	03/12/2000	Boyra	Khulna	Bangladesh	1990	01869844574	saikat@.com
P09	saikat	Biswas	03/12/2000	Boyra	Khulna	Bangladesh	1990	01957847834	saikat@.com
P11	Ruhan	-	08/09/2003	rampura dhaka	-	-	-	01390202030	-
P02	Shahriar	Ahmed	05/11/2001	Banasree	Dhaka	Bangladesh	1212	01845783456	sahriar@.com

More than 10 rows available. Increase rows selector to view more rows.

3. Insert data in Inventory table:

```
INSERT INTO Inventory (inventoryId, medicationId, quantity,expiryDate)
VALUES ( 'I01',  'M01',100,
    TO_DATE('2025-12-31', 'YYYY-MM-DD')
);
```

See demo data from Inventory table:

```
SELECT
   inventoryId,medicationId, quantity,expiryDate
FROM
   Inventory;
```

INVENTORYID	MEDICATIONID	QUANTITY	EXPIRYDATE
I08	M08	70	03/04/2022
I03	M03	90	12/23/2026
I10	M10	140	03/02/2026
I04	M04	150	04/22/2025
I02	M02	50	11/23/2024
I05	M05	130	07/28/2026
I07	M07	110	03/12/2025
I01	M01	100	12/31/2025
I06	M06	190	09/28/2024
I09	M09	90	09/04/2024

10 rows returned in 0.01 seconds Download

4. Insert demo data in Medication table:

```
INSERT INTO Medication (medicationId,name,description,manufacturer, price)
VALUES (
        'M01','Aspirin', 'Pain reliever and anti-inflammatory', 'Incepta',  5.90
);
```

See demo data from Medication table:

```
SELECT
 medicationId,name, description, manufacturer,price
FROM
   Medication;
```

MEDICATIONID	NAME	DESCRIPTION	MANUFACTURER	PRICE
M09	Optimox Myciten	Eye drop	square	390
M01	Aspirin	Pain reliever and anti-inflammatory	Incepta	5.9
M03	Etorix	Pain killer	Square	17
M04	Monas	Allergy reliver	Square	16
M05	Miolux	Pain killer	SKF	20
M06	Orbain	Blood pressure	SKF	200
M07	Surgel	Gastric medicine	square	90
M02	Napa	Headace and fever reliver	Incepta	6.9
M08	Esoral	Gastric medicine	Labaid	190
M10	Omidon	Gastric syrup	SKF	200

10 rows returned in 0.01 seconds Download

5. Insert demo data in OrderTable:

```
INSERT INTO OrderTable (orderId,patientId,orderDate,status, medications
) VALUES (
  'O01', 'P01',
  TO_DATE('2024-05-31', 'YYYY-MM-DD'), 'Processing',
  MedicationOrderList(
    MedicationOrder('M01', 2),
    MedicationOrder('M02', 3)
  )
);
```

See demo data from OrderTable:

```
SELECT
  o.orderId,o.patientId,o.orderDate,o.status,
  t1.medicationId,t1.quantity
FROM
  OrderTable o, TABLE(o.medications) t1;
```

ORDERID	PATIENTID	ORDERDATE	STATUS	MEDICATIONID	QUANTITY
O01	P01	05/31/2024	Processing	M01	2
O01	P01	05/31/2024	Processing	M02	3
O02	P02	04/03/2024	Delivered	M01	4
O02	P02	04/03/2024	Delivered	M02	30
O05	P05	06/23/2024	Pending	M10	14
O05	P05	06/23/2024	Pending	M09	20
O05	P05	06/23/2024	Pending	M08	20
O09	P09	04/08/2024	Cancelled	M05	15
O09	P09	04/08/2024	Cancelled	M08	24
O09	P09	04/08/2024	Cancelled	M02	34

More than 10 rows available. Increase rows selector to view more rows.

6. Insert demo data in Prescription table:

```
INSERT INTO Prescription ( prescriptionId,pdate, doctorId,patientId, medications
) VALUES (
   'P01', TO_DATE('2024-05-31', 'YYYY-MM-DD'), 'D02',  'P01',
   MedicationList(
      MedicationRef('M01', 14),
      MedicationRef('M04', 6)
   )
);
```

See demo data from Prescription table:

```
SELECT
   p.prescriptionId,p.pdate,p.doctorId,p.patientId,
   t.medicationId,t.quantity
FROM
   Prescription p,TABLE(p.medications) t;
```

PRESCRIPTIONID	PDATE	DOCTORID	PATIENTID	MEDICATIONID	QUANTITY
P05	05/31/2024	D05	P06	M03	16
P05	05/31/2024	D05	P06	M04	2
P05	05/31/2024	D05	P06	M06	24
P08	05/31/2024	D07	P08	M02	16
P08	05/31/2024	D07	P08	M01	22
P10	05/31/2024	D09	P10	M06	16
P10	05/31/2024	D09	P10	M09	22
P10	05/31/2024	D09	P10	M01	2
P10	05/31/2024	D09	P10	M10	12
P01	05/31/2024	D02	P01	M01	14

More than 10 rows available. Increase rows selector to view more rows.

Query:

-- 1. List all patients' full names and their contact email addresses

```
SELECT p.name.first_name || ' ' || p.name.last_name AS full_name,
      p.contactInfo.email
FROM Patient p;
```

```sql
--1. List all patients' full names and their contact email addresses
SELECT p.name.first_name || ' ' || p.name.last_name AS full_name,
       p.contactInfo.email
FROM Patient p;
```

Results | Explain | Describe | Saved SQL | History

FULL_NAME	
Anis Rahman	anisrahman@.com
Rakib Hosain	rakib@.com
Zafor Ahmed	zafor@.com
saikat Biswas	saikat@.com
Ruhan	-
Shahriar Ahmed	sahriar@.com
Arefin Ahmed	arefin@.com

--2. Find all medications with their stock quantity that have less than 90 units in stock.

SELECT mws.medicationId, mws.name, mws.stockQuantity
FROM MedicationWithStock mws
WHERE mws.stockQuantity < 90;

```sql
-- 2.Find all medications with their stock quantity that have less than 90 units in stock.
SELECT mws.medicationId, mws.name, mws.stockQuantity
FROM MedicationWithStock mws
WHERE mws.stockQuantity < 90;
```

Results | Explain | Describe | Saved SQL | History

MEDICATIONID	NAME	
M02	Napa	50
M08	Esoral	70

--3. Retrieve all orders made by a specific patient identified by patientId 'P03', including the total amount of each order.;

SELECT o.orderId, o.orderDate, o.status, owa.totalAmount
FROM OrderTable o
JOIN OrderWithTotalAmount owa ON o.orderId = owa.orderId
WHERE o.patientId = 'P03';

```sql
--3.Retrieve all orders made by a specific patient identified by patientId 'P03', including the total amount of each order.;
SELECT o.orderId, o.orderDate, o.status, owa.totalAmount
FROM OrderTable o
JOIN OrderWithTotalAmount owa ON o.orderId = owa.orderId
WHERE o.patientId = 'P03';
```

Results | Explain | Describe | Saved SQL | History

ORDERID	ORDERDATE	STATUS	TOTALAMOUNT
003	05/03/2024	Cancelled	4224

--4. Find all doctors who specialize in 'Cardiology' along with their contact phone numbers.

```sql
SELECT d.name.first_name || ' ' || d.name.last_name AS full_name,
       pn.COLUMN_VALUE AS phone_number
FROM Doctor d,
     TABLE(d.contactInfo.phoneNumbers) pn
WHERE d.specialization = 'Cardiology';
```

```sql
--4. Find all doctors who specialize in 'Cardiology' along with their contact phone numbers.
SELECT d.name.first_name || ' ' || d.name.last_name AS full_name,
       pn.COLUMN_VALUE AS phone_number
FROM Doctor d,
     TABLE(d.contactInfo.phoneNumbers) pn
WHERE d.specialization = 'Cardiology';
```

Results | Explain | Describe | Saved SQL | History

FULL_NAME	PHONE_NUMBER
Dr.Tuhin Shuvra	01783557895
Dr.Fazle Rabbi	01945782345

– 5.List all medications prescribed in a specific prescription identified by prescriptionId 'P04'

```sql
SELECT m.name, mo.quantity
FROM Prescription p
JOIN TABLE(p.medications) mo ON p.prescriptionId = 'P04'
JOIN Medication m ON mo.medicationId = m.medicationId;
```

```sql
--5.List all medications prescribed in a specific prescription identified by prescriptionId 'P04'

SELECT m.name, mo.quantity
FROM Prescription p
JOIN TABLE(p.medications) mo ON p.prescriptionId = 'P04'
JOIN Medication m ON mo.medicationId = m.medicationId;
```

Results | Explain | Describe | Saved SQL | History

NAME	QUANTITY
Optimox Myciten	14
Omidon	12
Surgel	12

--6. Calculate the total number of medications in the inventory and their combined total price

```sql
SELECT COUNT(*) AS total_medications,
       SUM(m.price * i.quantity) AS total_value
FROM Inventory i
JOIN Medication m ON i.medicationId = m.medicationId;
```

```sql
--6.Calculate the total number of medications in the inventory and their combined total price.Id = m.medicationId;
SELECT COUNT(*) AS total_medications,
       SUM(m.price * i.quantity) AS total_value
FROM Inventory i
JOIN Medication m ON i.medicationId = m.medicationId;
```

Results | Explain | Describe | Saved SQL | History

TOTAL_MEDICATIONS	TOTAL_VALUE
10	131765

– 7. Find the top 5 most expensive medications along with their price and manufacturer.

```sql
SELECT m.name, m.price, m.manufacturer
FROM Medication m
ORDER BY m.price DESC
FETCH FIRST 5 ROWS ONLY;
```

```sql
--7.Find the top 5 most expensive medications along with their price and manufacturer.
SELECT m.name, m.price, m.manufacturer
FROM Medication m
ORDER BY m.price DESC
FETCH FIRST 5 ROWS ONLY;
```

Results | Explain | Describe | Saved SQL | History

NAME	PRICE	MANUFACTURER
Optimox Myciten	390	square
Orbain	200	SKF
Omidon	200	SKF
Esoral	190	Labaid
Surgel	90	square

--8. Retrieve all patients who have not placed any orders.

SELECT p.patientId, p.name.first_name || ' ' || p.name.last_name AS full_name
FROM Patient p
LEFT JOIN OrderTable o ON p.patientId = o.patientId
WHERE o.orderId IS NULL;

```sql
--8.Retrieve all patients who have not placed any orders.
SELECT p.patientId, p.name.first_name || ' ' || p.name.last_name AS full_name
FROM Patient p
LEFT JOIN OrderTable o ON p.patientId = o.patientId
WHERE o.orderId IS NULL;
```

Results | Explain | Describe | Saved SQL | History

PATIENTID	FULL_NAME
P11	Ruhan

– 9. List all expired medications in the inventory.iryDate < SYSDATE;

SELECT m.name, i.expiryDate, i.quantity
FROM Inventory i
JOIN Medication m ON i.medicationId = m.medicationId
WHERE i.expiryDate < SYSDATE;

```sql
--9.List all expired medications in the inventory.iryDate < SYSDATE;
SELECT m.name, i.expiryDate, i.quantity
FROM Inventory i
JOIN Medication m ON i.medicationId = m.medicationId
WHERE i.expiryDate < SYSDATE;
```

Results | Explain | Describe | Saved SQL | History

NAME	EXPIRYDATE	QUANTITY
Esoral	03/04/2022	70

--10 .Generate a report of all orders, including the patient name, order date, status, and total amount

```
SELECT o.orderId,
      p.name.first_name || ' ' || p.name.last_name AS patient_name,
      o.orderDate, o.status, owa.totalAmount
FROM OrderTable o
JOIN Patient p ON o.patientId = p.patientId
JOIN OrderWithTotalAmount owa ON o.orderId = owa.orderId;
```

```
--10 Generate a report of all orders, including the patient name, order date, status, and total amount
SELECT o.orderId,
       p.name.first_name || ' ' || p.name.last_name AS patient_name,
       o.orderDate, o.status, owa.totalAmount
FROM OrderTable o
JOIN Patient p ON o.patientId = p.patientId
JOIN OrderWithTotalAmount owa ON o.orderId = owa.orderId;
```

Results | Explain | Describe | Saved SQL | History

ORDERID	PATIENT_NAME	ORDERDATE	STATUS	TOTALAMOUNT
O01	Anis Rahman	05/31/2024	Processing	32.5
O06	Rakib Hosain	06/17/2024	Pending	5260
O04	Zafor Ahmed	05/23/2024	Delivered	682
O09	saikat Biswas	04/08/2024	Cancelled	5094.6
O02	Shahriar Ahmed	04/03/2024	Delivered	230.6

Conclusion: The Pharmacy Management System DBMS Project was a good learning experience for implementing a real-world DBMS and helped us understand the nuances of full implementation. The most interesting part was the experience of starting from the ER diagram and then translating the concepts into the terms of a DBMS. It is capable of handling day-to-day operations for a pharmacy.